MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6AL7GT

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Electron-ray indicator for AM-FM receivers

Outline --- 3-4

Base --- B8-6

Envelope --- T-9

Cathode --- Coated unipotential

Base connections:

Pin No.	1	2	3	4	5	6	7	8
Element	--- g	h	ta	D2	D3	D1	h	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec	Rk	Eta	ED1	ED2	ED3	Ehk	RD1
Unit:	V	Vdc	Ohm	Vdc	Vdc	Vdc	Vdc	v	Meg
Maximum:	6.9	---	---	365	---	---	---	100	---
Minimum:	5.7	---	---	220	---	---	---	---	---
TEST CONDITIONS:	6.3	0	3300	300	0	0	0	---	1.0

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GENERAL:

Qualification - Not required

Method	REQUIREMENT OR TEST	CONDITIONS	AQL	Inspection Level or Code	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 1							
1266	Grid current	EDD1=EDD2=EDD3= -10Vdc; Ec=Eta= 67.5V; Rk=0 Notes 1 and 9	0.65	II	ID	0	-1.0	uAdc
1256	Target current		0.65	II	Ita	0.32	0.64	mAdc
---	Light output	ED1=ED2=ED3=10Vdc; RD1=0; Note 8	0.65	II		---	---	---
1231	Emission	Ec=ED1=ED2=ED3= 50Vdc; Eta=0; Rk=RD1=0 Note: 9	0.65	II	Is	15	---	mAdc
	Quality conformance inspection, part 2							
---	Grid cut-off	Ec=-7Vdc; Note 2	6.5	S3	IcCo			
---	Sensitivity (1)	Vary EDD1; Notes 3 and 5	2.5	I	EDD1	---	±5.0	Vdc
---	Sensitivity (2)	Vary EDD2; RD1=0; RD2=1.0 Meg; Notes 3 and 5	2.5	I	EDD2	---	±5.0	Vdc
---	Block out	EDD1=ED2=ED3=10Vdc; Eta=200Vdc; Note 6	2.5	I				
---	Pattern cutoff	ED1=ED2=ED3=-20Vdc; RD1=0; Note 7	2.5	I				
	Quality conformance inspection, part 3 Periodic-check test							
1301	Heater current		---	---	If	138	162	mA
1336	Heater-cathode leakage	Ehk=±100Vdc	---	---	Ihk	0	20	uAdc
---	Pattern check	Notes 3 and 4	---	---	EDD1	-0.5	+0.5	Vdc

NOTES:

1. D1, D2, and D3 shall be tied together through one 1-megohm resistor to -10Vdc. Read current in deflector circuit.
2. Pattern shall disappear.
3. The tube shall be mounted horizontally and a plane shall pass through pins 4 and 8 vertical with pin 4 on top.

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NOTES (Continued):

4. EDD1 shall be adjusted so that the top edge of the pattern forms a straight line. EDD1 shall be within the limits shown herein. Bottom edge of the pattern shall be visible.
5. Positive voltage shall be applied to the deflector plate to give an upward displacement of the top edge of the pattern of 1/16 inch and EDD shall be recorded. Polarity of EDD shall be reversed to give a downward displacement of the top end of the pattern of 1/16 inch and EDD shall be recorded. The deflector supply voltage necessary to produce the displacement of 1/16 inch in either direction shall be not greater than the limit specified herein. The pattern edge shall be reasonably sharp and the pattern area uniformly illuminated.
6. Target supply voltage of 200 Vdc shall be applied to the target across a 16 uf capacitor through 10,000 ohms. Pattern areas shall be free from non- fluorescent regions.
7. With the voltages specified herein applied to the deflector plates, the pattern shall not completely disappear.
8. Tubes having images which show dark areas or sponginess shall be rejected.
9. Test shall be performed at the conclusion of the holding period.

User Activities:
Navy-AS, OS, MC, CG, SH

Preparing Activity:
Navy-EC

(Project 5960-N851)

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DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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